Day 11 Problem Set

Geometry Summer Credit | Pythagorean Theorem, Converse, Distance Formula, and Radicals

Name:	Date:	Period:

Goal

Practice right-triangle side lengths, converse reasoning, coordinate distance, and simplifying radicals.

Support Practice

1.	Simplify: sqrt(36).

2.	Simplify: sqrt(45).

3.	Solve: x^2 = 169.

4.	A right triangle has legs 6 and 8. Find the hypotenuse.

5.	A right triangle has hypotenuse 10 and one leg 6. Find the missing leg.

Core Practice: Pythagorean Theorem

6.	Find the hypotenuse when the legs are 5 and 12.

7.	Find the hypotenuse when the legs are 8 and 15.

8.	Find the missing leg when the hypotenuse is 17 and one leg is 8.

9.	Find the missing leg when the hypotenuse is 20 and one leg is 12.

10.	A ladder is 13 feet long and reaches a window 12 feet high. How far is the base of the ladder from the wall?

11.	A rectangle has side lengths 9 and 12. Find the length of its diagonal.

Day 11 Problem Set

Continued

Core Practice: Converse and Coordinate Distance

12.	Do 9, 12, and 15 form a right triangle? Show the square comparison.

13.	Do 7, 24, and 25 form a right triangle? Show the square comparison.

14.	Do 5, 8, and 10 form a right triangle? Show the square comparison.

15.	Find the distance between A(0, 0) and B(9, 12).

16.	Find the distance between C(-1, 2) and D(7, 8).

17.	Find the distance between E(-4, -3) and F(2, 5).

18.	Find the distance between G(1, -6) and H(8, -2). Leave your answer in simplest radical form.

19.	Simplify: sqrt(75).

20.	Simplify: sqrt(128).

21.	Spiral review: Triangle ABC is similar to triangle DEF. AB = 10, DE = 25, BC = 14. Find EF.

22.	Challenge: A student says the distance between (0,0) and (6,8) is sqrt(6^2) + sqrt(8^2) = 14. Explain and correct the mistake.